problems/MHD/rotorDaoNazarov2022

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. Read off `user_inputs.jl`, the case files and the kernels they reach on `sm/MHD`. The case directory already carries `algorithm.tex`/`algorithm.pdf`, a full derivation of this deck's algorithm (including the Runge–Kutta coefficients); this note restates the DynSGS and positivity part in the common format of `problems/MHD/setup_notes` so that it can be compared with the other cases, and was checked against the code independently of that document (they agree).

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	2D ideal GLM-MHD, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$	the Orszag–Tang flux with $\gamma = 1.4$
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2 - \frac{1}{2}\psi^2)$, $\gamma = 1.4$	no flux floor
Domain, mesh	$[0,1]^2$, doubly periodic; `OT_32x32_periodic.msh` refined twice by p4est $\Rightarrow 128 \times 128$ quads	`:init_refine_lvl => 2`
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 3$ (385^2 nodes)	
Time integration	`CarpenterKennedy2N54` (rebuilt with the limiter, Section 5), fixed $\Delta t = 2 \times 10^{-4}$, $t_{end} = 0.15$	
Variables	`TOTAL()`, `"energy"`	
Initial data	ambient $(\rho, \mathbf{v}, p, \mathbf{B}) = (1, 0, 1, (5/\sqrt{4\pi}, 0))$; disc $r < 0.1$: $\rho = 10$, $\mathbf{v} = (u_0/r_0)(0.5 - y,\ x - 0.5)$; taper $0.1 \le r < 0.115$: $\rho = 1 + 9f$, $\mathbf{v} = (f u_0/r)(\dots)$, $f = (r_1 - r)/(r_1 - r_0)$, $u_0 = 2$	`initialize.jl`
GLM cleaning	$c_h = \max(\lvert\mathbf{v}\rvert + c_f)$ of the initial data, constant; $S_\psi = -(c_h/0.18)\psi$	
Sources	`:lsource => true` (ψ damping only)	
Diagnostics	mass and energy totals every step (`:conservation_every => 1`, slots 1 and 4)	`conservation.dat`
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false	

2 DynSGS: deck keys and effective values

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`, `:lvisc => true`	`SGS.jl:1598`
$:\mu$	all 1	
`:dsgs_sensor`	`"legacy"`	
`:dsgs_CR`, `:dsgs_Cmax`, `:dsgs_Cmin`	1.0, 0.5, 0 (default)	
`:dsgs_gamma`, `:dsgs_Prt`	$= \gamma_{mhd} = 1.4$, 1.0	
`:dsgs_norms`, `:dsgs_local_rel`	`"element"`, 1.0 (default)	
`:dsgs_rel`	1.0	unused with element norms
`:dsgs_conserved`	true	
`:dsgs_nazarov_energy`	false	deck: with `true`, p dipped to -0.02 at 64^2
`:dsgs_fast_floors`, `:dsgs_thermal_residual`, `:dsgs_ref_weight`	false (defaults)	
`:dsgs_hold_steps`	2	
`:dsgs_cutoff`, `:dsgs_freeze_stage`, `:dsgs_reference`, `:dsgs_nodal_rho`, `:ldsgs_nodal`	0, false, false, false, false	defaults

3 DynSGS as it runs for this case

$\Delta_K^{elem} = 1/128$, $\Delta_K = \Delta_K^{elem}/(N+1) = 1/512$, $h = \Delta t$.

1. Rotation once per step; two held steps ($\nu = 0$).
2. **Legacy residual**, every stage: $R_I^s = \left|\frac{3q_I^s - 4q_I^{n,s} + q_I^{n-1,s}}{2h} - (M^{-1}\mathcal{R}_{inv})_I^s\right|$, $\mathcal{R}_{inv}$ assembled over ranks and periodic copies; the state q is the already-limited stage state.
3. **Element normalization**: $n_K^s = \max(S_K^s, \sigma_K^s) + 10^{-16}$ over the 16 nodes of K, with $\sigma_K = (\bar\rho_K,\ \bar\rho_K \bar c_K,\ \bar\rho_K \bar c_K^2,\ \sqrt{\bar\rho_K}\bar c_K)$ for $(\rho,\ \mathbf{m},\ E,\ \mathbf{B})$, $\bar c_K = \sqrt{\gamma \bar p_K/\bar\rho_K}$ (sound speed).
4. $\mathcal{R}_K = \max_{I\in K}\max_{s<8} R_I^s/n_K^s$, $w_K = \max_I(|\mathbf{v}_I| + \sqrt{\max(\gamma p_I/\rho_I + |\mathbf{B}_I|^2/\rho_I, 0)})$, $\nu_K = \max(0, \min(0.5\,\Delta_K w_K,\ \Delta_K^2 \mathcal{R}_K))$; no floor, no cutoff; one value per element, recomputed at every stage; ν_K on all nine slots.

4 The viscous operator

`user_primitives.jl` copies $\mathbf{q}$ (the energy split is coded but inactive, `:dsgs_nazarov_energy => false`), so

$$\partial_t \mathbf{q} + \nabla\cdot\mathbf{F}(\mathbf{q}) = \nabla\cdot(\nu_K \nabla\mathbf{q}) \qquad \text{on } (\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi),$$

a Laplacian of the conserved state, mass diffusion on, no $\tau\cdot\mathbf{v}$ or resistive-work term.

5 Positivity: conservative element limiter

Deck key / item	Value as run	Source / remark
`:lpositivity`	**true**	
`:positivity_method`	`"conservative"`	
`:positivity_rho_min`	10^{-6} $(= 10^{-6}\times$ ambient $\rho)$	
`:positivity_p_min`	10^{-6} $(= 10^{-6}\times$ ambient $p)$	

Deck key / item	Value as run	Source / remark
`:positivity_report`, `_every`	true, 1000 (defaults)	1000 limiter calls = 200 steps

How the limiter is hooked in

- `:lpositivity => true` and `:positivity_method => "conservative"` make `mod_inputs.jl:836-838` rebuild the deck's `CarpenterKennedy2N54` with the limiter as its stage limiter and `williamson_condition = false`:

  ```
  CarpenterKennedy2N54(stage_limiter!  = positivity_conservative_limiter!, williamson_condition =
                                                                  false)
  ```

  (`positivity_driver.jl:263`); any other solver, or one that already has a stage limiter, is refused.
- The limiter therefore runs before the right-hand side of stages 2–5 and once after stage 5: five times per step, on every rank. The state entering stage 1 is the limited end of the previous step; the initial condition itself is not limited.
- On the first call `_cpos_validate` (`positivity_driver.jl:271`) requires: the nine-field `qvars` $(\rho, \rho u, \rho v, \rho E, \rho w, B_x, B_y, B_z, \psi)$, 2D, continuous Galerkin, a lumped (vector) mass matrix, `TOTAL()`, CPU, no `:ladapt`, `:dsgs_gamma` set, both floors > 0. γ is taken from `:dsgs_gamma`, not from `PhysicalConst`.
- The node-wise *repair* of `apply_positivity!` (`rhs.jl:821`) returns at once in this mode: nothing is done inside `rhs!`.

What one call does

Let $\tilde{E} = E - \frac{1}{2}|\mathbf{B}|^2 - \frac{1}{2}\psi^2$ (the part of E the limiter may touch), $p(\rho, \mathbf{m}, \tilde{E}) = (\gamma-1)(\tilde{E} - |\mathbf{m}|^2/(2\rho))$, $\varepsilon_\rho = 10^{-6}$, $\varepsilon_p = 10^{-6}$, $s = 64$, and $\mathrm{ulp}(x)$ the double-precision spacing at x.

1. **Mark** (`cpos_mark_bad!`): node I is bad unless $\rho_I \geq \varepsilon_\rho$ and $p_I \geq \varepsilon_p$ (a NaN node is therefore bad). The number of bad nodes is `MPI.Allreduce`d; if it is 0 everywhere the call returns and the state is untouched.
2. **Element pass** (`cpos_element_pass!`), for every element K holding a bad node, with the element mass weights $w_I = \omega_i \omega_j J_{K,ij}$, $W = \sum_I w_I$:
   - means $\bar{\rho} = \sum w_I \rho_I / W$, $\bar{\mathbf{m}}$, $\bar{E}$ (of $\tilde{E}$); round-off margins $\delta_\rho = s\,\mathrm{ulp}(\max(\max_I|\rho_I|, |\bar{\rho}|))$, $\bar{\rho}_g = \bar{\rho} - \delta_\rho$, $\varepsilon_\rho^+ = \varepsilon_\rho + \delta_\rho$, $\varepsilon_p^+ = \varepsilon_p + s(\gamma-1)\,\mathrm{ulp}(\max_I(|E_I| + \frac{1}{2}|\mathbf{B}_I|^2 + \frac{1}{2}\psi_I^2))$;
   - if the mean is not admissible ($\bar{\rho}_g \leq \varepsilon_\rho^+$ or $p(\bar{\rho}_g, \bar{\mathbf{m}}, \bar{E}) \leq \varepsilon_p^+$, or not finite) the element is counted as *failed* and skipped;
   - density: if $\min_I \rho_I < \varepsilon_\rho^+$, $\theta_\rho = \mathrm{clamp}\big((\bar{\rho} - \varepsilon_\rho^+)/(\bar{\rho} - \min_I \rho_I), 0, 1\big)$, else $\theta_\rho = 1$; $\tilde{\rho}_I = \bar{\rho} + \theta_\rho(\rho_I - \bar{\rho})$;
   - pressure: for every node with $p(\tilde{\rho}_I - \mathrm{margin}, \mathbf{m}_I, \tilde{E}_I) < \varepsilon_p^+$, $t_I \in [0,1]$ is the smallest root of the quadratic $(\bar{\rho}_g + t\,d\rho)(\bar{E} - e^+ + t\,dE) - \frac{1}{2}|\bar{\mathbf{m}} + t\,d\mathbf{m}|^2 = 0$, $e^+ = \varepsilon_p^+/(\gamma-1)$, $d\rho = \tilde{\rho}_{I,g} - \bar{\rho}_g$, $d\mathbf{m} = \mathbf{m}_I - \bar{\mathbf{m}}$, $dE = \tilde{E}_I - \bar{E}$ (computed with the cancellation-free root formula); $\theta_p = \min_I t_I$ ($\theta_p = 0$ if some $\tilde{\rho}_{I,g} \leq 0$);
   - increments, accumulated as $w_I\,\delta q_I$:

     $$\delta\rho_I = (\theta_p\theta_\rho - 1)(\rho_I - \bar{\rho}), \quad \delta\mathbf{m}_I = (\theta_p - 1)(\mathbf{m}_I - \bar{\mathbf{m}}), \quad \delta E_I = (\theta_p - 1)(\tilde{E}_I - \bar{E}), \quad \delta\mathbf{B}_I = \delta\psi_I = 0.$$

     Since $\bar{q}$ is the w-weighted mean, $\sum_I w_I \delta q_I = 0$ in every element: ρ, $\rho\mathbf{v}$ and E are conserved exactly; $\mathbf{B}$ and ψ (hence $\nabla\cdot\mathbf{B}$) are never touched.
3. **Assemble and apply** (`DSS_global_RHS!`, `cpos_add!`): the element increments are summed over elements, ranks and periodic copies and divided by the lumped global mass, $q_I \leftarrow q_I + M_I^{-1}\sum_K w_I^K \delta q_I^K$. At a node shared by several elements the update is the mass-weighted average of their increments, so the assembled node need not reproduce the element's limited value exactly.

4. **Node floor** (`cpos_floor!`, not conservative), for the finite nodes still out of bounds (e.g. in failed elements): $\rho \leftarrow \varepsilon_\rho + s\,\mathrm{ulp}(\varepsilon_\rho)$ if $\rho < \varepsilon_\rho$ (mass added); then with $k = |\mathbf{m}|^2/(2\rho)$ and $e^+ = \varepsilon_p/(\gamma-1) + s\,\mathrm{ulp}(|E| + \frac{1}{2}|\mathbf{B}|^2 + \frac{1}{2}\psi^2)$, if $\tilde{E} - k < e^+$: either scale $\mathbf{m}$ by $\sqrt{(\tilde{E} - e^+)/k}$ (E unchanged) when $\tilde{E} > e^+$, or set $\mathbf{m} = 0$ and raise E by $e^+ - \tilde{E}$ (energy added).
5. **Report** (`_cpos_report`): every 1000 limiter calls (`:positivity_report_every`), when the global counts have grown: element limitings, $\min\theta_\rho$, $\min\theta_p$, failed elements, node floors, and in how many calls the limiter was active.

6 Observations

1. Deck, comments and `algorithm.tex` agree with the code. The deck sets no flux-only floor; a non-admissible state can only reach the flux if the limiter's node floor skips it (non-finite values).

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	Where (file:line)
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report	`src/io/mod_inputs.jl:456-477`
`:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2,H)+1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau\cdot\mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau\cdot\mathbf{v}$ 3076, $\eta\,\mathbf{B}\cdot\nabla\mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs}+2$ in 2D, $n_{eqs}+1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)